ADAPTIQ — ADAPTIVE INTELLIGENCE FOR PERSONALIZED LEARNING

[123] CSE

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PART A: 2-MARK QUESTIONS & VERIFIED ANSWERS

Concise Definitions, Key Invariants, Operational Differences & Formulas

Total Questions: 7 • Marks per Question: 2 Marks • Total Marks: 14 Marks

INDEX OF COVERED TOPICS & LEARNING UNITS:

1. Power Flow Diagram • 2. Free Space Loss • 3. Non-linearity in HPA • 4. Bent-Pipe Transponder • 5. Link Budget Analysis • 6. Satellite Interference • 7. Numerical Problem

QUESTION 1 Topic: Power Flow Diagram **[2 Marks]**

Q1. State the purpose of an uplink in a satellite communication system.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: sc unit 3 (Page 6) — "Information from the transmitting earth station is sent to the satellite through the uplink...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The uplink is the communication link used to send information from the transmitting earth station to the satellite. It forms the first major stage of the satellite transmission path before the signal is processed and retransmitted.

QUESTION 2 Topic: Free Space Loss **[2 Marks]**

Q2. Define Free Space Transmission Loss.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: sc unit 3 (Page 5) — "Free space transmission loss is the loss in signal power as the electromagnetic wave propagates through space ..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Free space transmission loss is defined as the loss in signal power as the electromagnetic wave propagates through free space without encountering any physical medium.

QUESTION 3 Topic: Non-linearity in HPA **[2 Marks]**

Q3. Define Output Back-Off (OBO) in the context of high-power amplifiers.

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: sc unit 3 (Page 4) — "Output Back-Off (OBO) is the amount by which the actual operating output power is kept below the saturation po..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Output Back-Off (OBO) is the specific amount by which the actual operating output power is kept below the saturation power of the amplifier to prevent non-linear distortion.

QUESTION 4 Topic: Bent-Pipe Transponder **[2 Marks]**

Q4. How is the combined Carrier-to-Noise (C/N) ratio of a bent-pipe transponder satellite link calculated?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: sc unit 3 (Page 3) — "Hence, the combined C/N ratio of a bent-pipe transponder satellite link is given by the inverse sum formula...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The combined C/N ratio of a bent-pipe transponder satellite link is determined using the inverse sum formula, combining the individual uplink and downlink C/N contributions.

QUESTION 5 Topic: Link Budget Analysis **[2 Marks]**

Q5. Write the formula for received carrier power (C) in a satellite link budget analysis and define its parameters.

Difficulty: HARD | Bloom: Remember

[SOURCE GROUNDING]: sc unit 3 (Page 2) — "C = P_T G_T G_R \left(\frac{\lambda}{4\pi R} \right)^2 \frac{1}{L_T L_R L_s L_{atm}}..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The received carrier power is given by C = P_T G_T G_R (lambda / 4 pi R)^2 * (1 / (L_T L_R L_s L_atm)), where P_T is transmitter power, G_T and G_R are antenna gains, lambda is wavelength, R is range, and L terms represent various system and atmospheric losses.

QUESTION 6 Topic: Satellite Interference **[2 Marks]**

Q6. What causes satellite circuit interference when multiple transponders are utilized?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: sc unit 3 (Page 1) — "When many satellite circuits (transponders) are used in the same frequency band, the signal from one circuit m..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

When many satellite circuits or transponders are operated within the same frequency band, interference occurs because the signal from one circuit may bleed into or disrupt another circuit.

QUESTION 7 Topic: Numerical Problem **[2 Marks]**

Q7. Identify the transmission specifications of the C-band transponder carried by a geostationary satellite from the numerical problem setup.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: sc unit 3 (Page 7) — "A geostationary satellite carries a C-band transponder which transmits 20 watts into an antenna with an on-axi..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The geostationary satellite carries a C-band transponder that transmits an output power of 20 watts into an antenna possessing an on-axis gain of 30 dB.